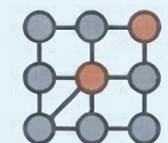
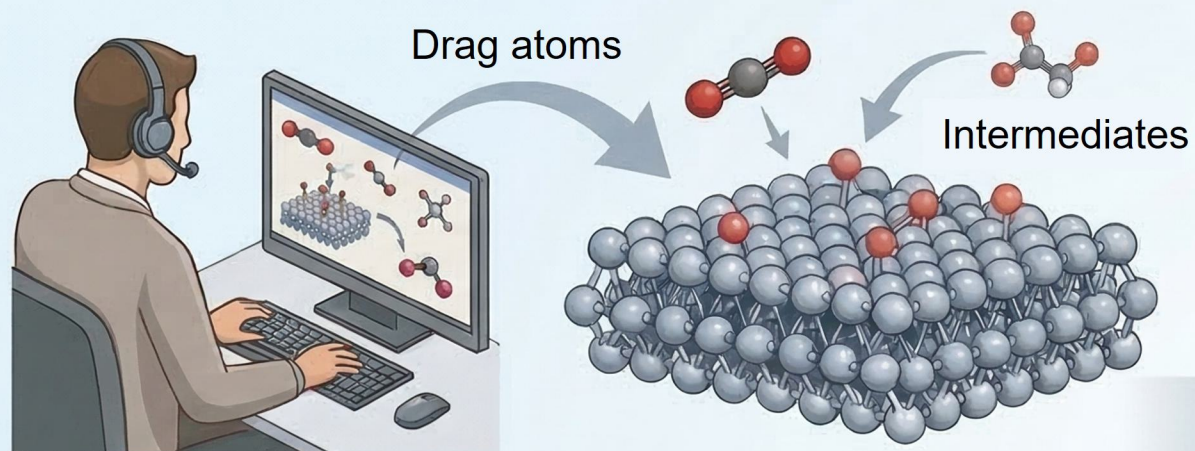


TRADITIONAL MANUAL CATALYSIS THEORY

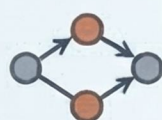
The Limitation: Time & Assumptions



Manual
structure
building



VASP
optimization



NEB
calculation



Slow growth
calculation



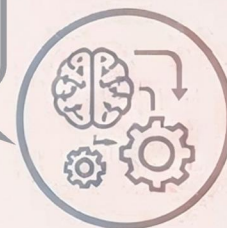
6-30 hours
PER CATALYST

- ✗ Static surface
- ✗ Idealized facets only
- ✗ Clean/fixed coverage
- ✗ No electric field
- ✗ Implicit pH

CATALYSIS DIGITAL TWIN

The Advantage: Rapid, Realistic & AI-Driven

Calculate CO₂ to
CH₄ with bulk.cif



Multiple
facets simulation



Dynamic
reconstruction



Realistic
potentials & fields



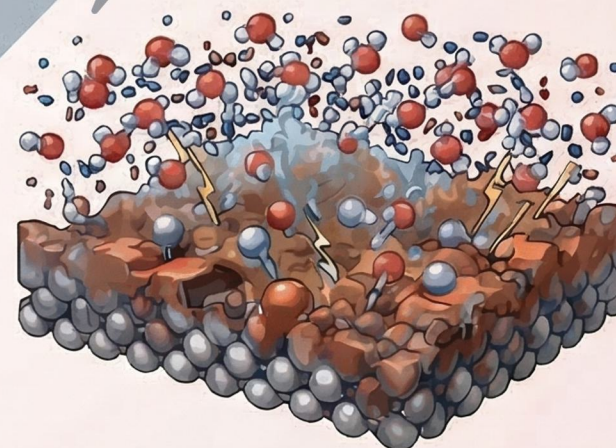
Multi-pathway
exploration



Accurate
predictions

5-30 mins

No human intervention



**ADDRESS ALL LIMITATIONS &
DELIVERS REAL-WORLD RESULTS**